

Project Executable Files

Date	8 July 2026
Team ID	xxxxxxx
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example similar) or	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

SmartLender1/
├── app.py
├── train_model.py
├── model.pkl
├── scaler.pkl
├── history.csv
├── requirements.txt
├── templates/
│   ├── home.html
│   ├── predict.html
│   ├── result.html
│   └── history.html
├── static/
│   └── style.css
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	<i>[Insert your IBM Cloud or hosting application URL here]</i>
Login Credentials (Demo)	No authentication required / Publicly accessible portal
Platform / Hosting Provider	IBM Cloud
Repository Link	<i>[Insert your GitHub Repository URL here]</i>
Demo Video Link	<i>[Insert your Walkthrough video link here]</i>

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Step	Action / Command	Description / Purpose
Pre-requisites	• OS: Windows 10/11, Linux, or macOS	Ensure your local system meets the technical development requirements before initialization.

Step	Action / Command	Description / Purpose
	• Environment: Anaconda Navigator / Python 3.8+ • IDE: VS Code / PyCharm • Browser: Google Chrome / Microsoft Edge	
Step 1	cd Smart-Lender	Open your terminal, Command Prompt, or PowerShell and navigate directly into the root project directory.
Step 2	pip install -r requirements.txt	Run this package command to download and install all necessary framework environment dependencies automatically.
Step 3	[Optional Check]	Configure environment parameters or verify baseline data transformation/mapping utilities housed inside utils.py .
Step 4	python app.py (Setup)	Verify model configurations or execute any required script migrations prior to deployment hosting.
Step 5	python app.py (Launch)	Start the application gateway using the production-ready Waitress WSGI engine to eliminate development server warnings.

Step	Action / Command	Description / Purpose
Step 6	Open browser to http://localhost:5000	Open your preferred browser (Chrome/Edge) and go to the local host address to interact with the Smart Lender portal.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Flask development server throws security and stability warnings in the standard terminal console.	Resolved: Upgraded the application backend codebase to deploy reliably using the production-ready Waitress WSGI server engine (serve()).
2	Feature shape mismatch or scaling errors when feeding raw web inputs directly into the XGBoost classifier pipeline.	Handled: Modularized all user input preprocessing routines into utils.py, leveraging a saved scaler.pkl matrix to normalize features accurately before evaluation.
3	Web application crashes entirely if the core serialized model file (model.pkl) is missing upon server initialization.	Handled: Embedded safety try-except catch routines to render a friendly error UI view rather than crashing, prompting the evaluator to execute train_model.py.